

Indian Institute of Technology Gandhinagar

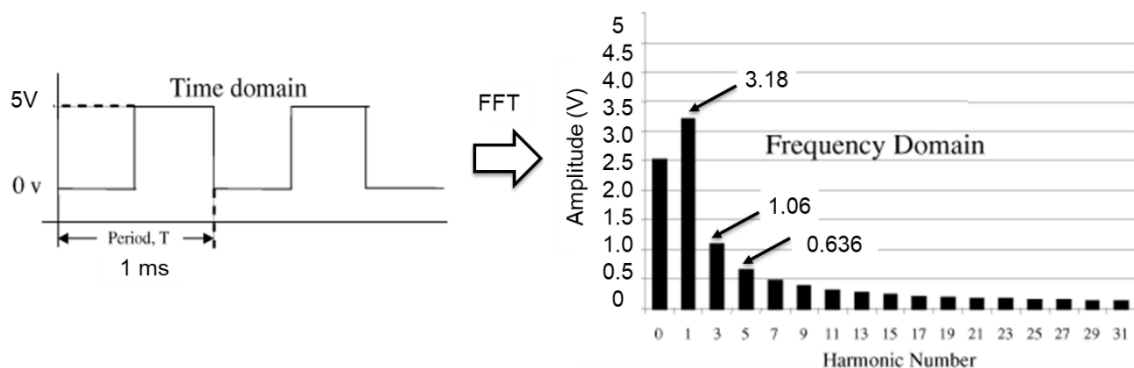
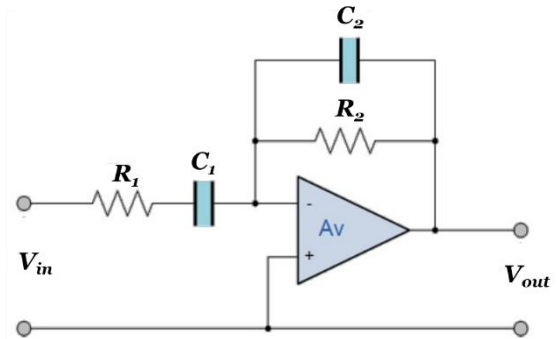
ES116 Principles and Applications of Electrical Engineering

Semester II, 2025-2026

Tutorial-9 (Interfacing Analog and Digital Worlds)

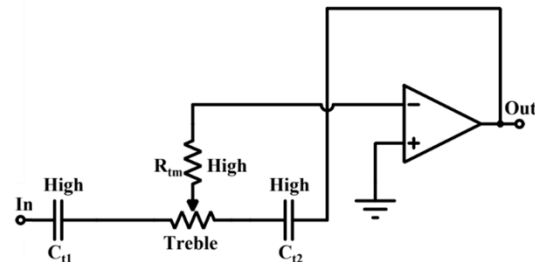
Q1. You are designing an analog input stage for a digital voice recorder with the given circuit ($R_1 = 1 \text{ k}\Omega$, $C_1 = 1 \text{ }\mu\text{F}$, $R_2 = 10 \text{ k}\Omega$, $C_2 = 10 \text{ nF}$). The raw microphone signal (V_{in}) picks up the user's voice, but it also accidentally picks up a 60 Hz low-frequency electrical hum and a 5 kHz high-frequency hiss.

- Calculate the frequency response of the given circuit (V_{out}/V_{in} with frequency).
- Explain exactly which of the three frequencies (60 Hz, 1 kHz, or 5 kHz) will successfully pass through to V_{out} and why.
- Imagine you feed a 1 kHz digital square wave (bouncing from 0 V to 5 V) into this circuit (frequency domain plot given). Based on the obtained frequency response, explain two specific ways the shape and position of this wave will change at the output.



Q2. You are analyzing the Treble Control circuit shown here, which uses a standard 741 Op-Amp (Open loop Gain = 107.7 dB and $f_{3dB} = 4.1 \text{ Hz}$). The goal of this circuit is to boost high-frequency audio from 4 kHz to 20 kHz.

- At the Maximum Boost setting, the potentiometer divides such that the input resistance is 1 k Ω and the feedback resistance is 100 k Ω . Calculate the high-frequency voltage gain magnitude. You can assume capacitors to be short-circuited.
- Find the Unity-Gain Bandwidth (f_t) for 741 Op-Amp.
- Using the obtained gain and f_t , calculate the usable cut-off frequency (f_{-3dB}) of this amplifier.

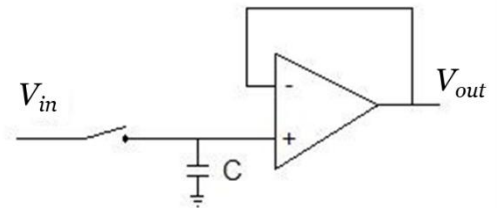


- D. Based on your calculated cut-off frequency, will this amplifier successfully boost the entire 4 kHz to 20 kHz audio range? Briefly explain why or why not.

Q3. You are designing a digital thermometer. You have an analog temperature sensor that outputs exactly 0 V at 0°C and 5 V at 100°C (it is perfectly linear). You feed this sensor into a basic 4-bit ADC that uses a 5 V reference.

- Calculate the voltage size of one Least Significant Bit (LSB) for this 4-bit ADC.
- Convert this LSB voltage into a temperature value. What is the smallest physical change in temperature this system is capable of measuring?
- The room temperature slowly and smoothly rises from 20°C to 24°C. Describe exactly what digital values the microcontroller sees during this time. Does it know the room is getting warmer?

Q4. You are analyzing the Sample and Hold circuit shown in the diagram. V_{in} is a 1 kHz sine wave with a peak amplitude of 5 V. The switch is controlled by a digital clock that momentarily closes ($\sim 1 \mu s$) to take a sample at a rate of 8 kHz. The switch has an ON resistance of 100 Ω and an OFF resistance of 10 M Ω . The holding capacitor (C) is 1 nF.



- Calculate how many samples are taken during exactly one full cycle of the 1 kHz sine wave. On a set of axes, sketch one full cycle of V_{in} and superimpose the resulting V_{out} waveform.
- Calculate the charging time constant (τ) when the switch is closed (ON). Is this fast enough to capture the voltage instantly?
- During the Hold phase, the switch is open. If you removed the Op-Amp buffer and connected the capacitor directly to an ADC that has a low input resistance of 1 k Ω , what would happen to the held voltage?

Q5. You are analyzing the 4-bit DAC shown in the schematic. A microcontroller provides the digital inputs (V_1 , V_2 , V_3 and V_4). A logic HIGH is 5 V and a logic LOW is 0 V. The feedback resistor (R_f) is chosen to be equal to $2R$.

- Based purely on how the resistors are arranged in this schematic, which input voltage (V_1 or V_4) represents the MSB, and which represents the LSB? Explain why?
- Assuming an ideal Op-Amp, write the full equation for the output voltage (V_o) in terms of the input voltages.
- The microcontroller sends the binary word 1010 to the DAC. Calculate the exact analog output voltage produced by this digital input.

